

UID → 25MC110036
Name → Nikhil Kumar
class → 25MAN-1(A)

classmate
Date _____
Page _____

Front End Assignment

Q.7) Make a redux app to make a Counter which will start from 0. It contains 2 buttons with increment and decrement options and count increases or decreases on the basis of which button you have clicked.

Ans Redux is a state management library used mainly with React to manage applications data in a predictable way.

Key Ideas:

- * Store → Central place where all state lives.
- * Action → Plain object describing what happened (eg increment)
- * Reducer → Function that updates state based on action.
- * Dispatch → Method to send actions to reducer.

Flow:

Component → dispatch(action) → reducer → updated state → UI re-renders.

Redux helps when multiple Components need shared state or when state logic becomes complex.

Basic objectives

Create a Redux based app 'Counter':

- Initial app: 0 (value)
- Two buttons: Increment (+), Decrement (-)
- Count updates based on button clicks.

Concepts Used :

Redux Toolkit (@reduxjs/toolkit)

React - Redux (useSelector, useDispatch)

Centralized state Management.

steps to make the app :

1. > Install dependencies :

```
npm install @reduxjs/toolkit react-redux
```

2. > Create Redux store :

```
import { configureStore } from '@reduxjs/toolkit';  
import CounterReducer from '../features/counter/CounterSlice';  
export const store = configureStore({  
  reducers: {  
    counter: CounterReducer,  
  }  
});
```

3. > Create Counter slice :

```
import { createSlice } from '@reduxjs/toolkit';  
const initialState = {  
  value: 0;  
};  
const CounterSlice = createSlice({  
  name: 'counter',  
  initialState,  
  reducers: {  
    increment: (state) => {  
      state.value += 1;  
    }  
  }  
});
```



```

    decrement: (state) => {
      state.value -= 1;
    },
  },
};

```

```

export const { increment, decrement } = counterSlice.actions;
export default counterSlice.reducer;

```

4. > Create Counter Component

```

import React from 'react';
import { useSelector, useDispatch } from 'react-redux';
import { increment, decrement } from './counterSlice';
const Counter = () => {
  const count = useSelector((state) => state.counter.value);
  const dispatch = useDispatch();
  return (
    <div style = {textAlign: 'center'} >
      <h1> Counter: {count} </h1>
      <button onClick = {() => dispatch(increment())}>
        Increment </button>
      <button onClick = {() => dispatch(decrement())}>
        Decrement </button>
    </div>
  );
};
export default Counter;

```

5. > Connect Redux to app:

```

import React from 'react';
import Counter from './features/Counter/Counter';
import ReactDOM from 'react-dom/client';

```



```

import App from './App';
ReactDOM.render(
  <Provider store={store}>
    <App/>
  </Provider>
);

```

6. > Use Counter in App:-

```

import Counter from './Counter';
function App() {
  return (
    <div>
      <Counter/>
    </div>
  );
}

```

* Flow diagram

